

Focus on research

Little Albert

What happened in Watson and Rayner's classic experiment with Little Albert? Watson was investigating how specific fears might be conditioned. His hypothesis was that a child would react in fear when they heard a loud noise, and that this fear was an innate (inborn) or unconditioned response. Therefore, according to classical conditioning theory, he could condition a child to have a fear response to other stimuli which normally would not be feared by a child. With this experimental design in mind, Watson and Rayner chose as their participant the child of a worker at the clinic in which the experiment would take place. It is believed that Little Albert's mother was unaware that her son would be used in an experiment on conditioning fear responses. As there was no informed consent in 1920, this is one of the reasons why this experiment is now considered unethical and would not be conducted today (see Chapter 10, 'Ethical considerations in psychological research' on page 347). Albert was considered suitable for the experiment as he was a placid child who rarely cried or seemed sad; thus, his stability was one of the main reasons for using him as a participant. Watson also felt that relatively little harm could be done to Albert.



Figure 9.16 Watson and Rayner's experiment with 11-month-old 'Little Albert'. Here, they are testing Albert for a fear response by presenting him with a scary mask.

To determine that Albert was capable of a fear response (UCR), Watson and Rayner pre-tested him and found that he was (see Figure 9.16). And so the fun began (but probably not for Albert!). For the experiment, Albert was placed on the floor in the middle of the room, and a white laboratory rat (NS) was placed nearby. Albert was allowed to play with the rat and showed no fear of the rat at this stage. In fact, he looked like he was rather enjoying himself.

This procedure was repeated, and then the next stage of the experiment was conducted. Whenever Little Albert reached out to touch the rat (NS), a steel bar behind Albert was struck with a hammer, causing a loud noise (UCS). The noise produced by the steel bar would have been loud enough to startle most people. Albert reacted by starting violently, his breathing

became irregular and he raised his arms. On the second instance the bar was struck, Albert had the same reaction, but this time his lips began to pucker and tremble. The third time, Albert started to cry. It was the first time an emotional situation caused a fear response in Albert. After several repetitions of this step, Albert would burst into tears and tremble (CR) at the mere sight of the rat (CS), and would become rather distraught. Clearly, Albert was conditioned: he had built up an association between the rat and an unpleasant stimulus (loud noise).

But they were not finished there! Five days later, Little Albert showed stimulus generalisation of his conditioning, as he reacted with fear when presented with a white rabbit, a fluffy dog, and a sealskin coat, not to mention some of his toys that were similar to the rat (Myers, 2001). At this stage, Albert's mother discovered what they were doing to her child, quit her job, and left the area. Watson and Rayner could not reverse (extinguish) the conditioning, therefore poor Albert was probably left with an irrational fear or phobia of anything white and fluffy (Garcia-Ives, Marquez & Perri, 1999). Psychologists have had little success in replicating or repeating the findings of Watson and Rayner with other children. The experiment conducted by Watson is an example of a case where ethical considerations were not followed as they would have to be today.

There are a number of reasons why Watson and Rayner's experiment is now considered to be unethical. The experiment was designed to condition an emotional response of fear in the participant. Through this conditioning it could be reasonably assumed that Albert was perhaps emotionally traumatised, and that he may have suffered lasting psychological harm from participating in the experiment. Watson failed to properly seek permission from Albert's mother (guardian), therefore no informed consent was obtained, and he did not debrief either Albert or his mother, to extinguish the conditioned response of fear to white rats and like objects. Who knows what happened to Little Albert? It is hoped he overcame the fear he had acquired of white objects!

Questions

- 1 What was the aim of the experiment?
- 2 Why did they choose Albert as their subject?
- 3 Briefly outline the procedure of the experiment.
- 4 Identify the following: NS, UCS, UCR, CS, CR.
- 5 What conclusions did Watson draw from the experiment?
- 6 How was stimulus generalisation used in the experiment?
- 7 Identify and explain the ethical problems involved in the experiment.
- 8 Explain how classical conditioning principles could be used to extinguish the conditioned response Albert developed.